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Professionalism in the Development of the FINAL Prototype

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Professionalism in the Development of the FINAL Prototype

1:Professionalism in the FINAL Prototype of SOULSTRIVENET

1.1 Introduction

This companion report evaluates the professionalism aspects of the SOULSTRIVENET project, focusing on its social impact, ethical considerations, legal implications, and security responsibilities. SOULSTRIVENET is an AI-enhanced social fitness platform that integrates a social feed with an AI chatbot ("FitBot") designed to provide personalized fitness guidance through prompt-driven interactions.

Given that the system handles personal user data, health-related preferences, and AI-generated advice, it raises important professional responsibilities aligned with industry best practices and UK/EU regulations. This report critically examines how these responsibilities were considered during design and development, supported by relevant academic and legal sources.

1.2 LITRATURE REVIEW

The development of the early prototype for STRIVENET demonstrates professionalism through ethical design, user-focused decision-making, structured development practices, and attention to real-world social, physical, and psychological impact. Even at this preliminary stage consisting of a landing page, registration interface, and login screen the artefact reflects responsibility, inclusivity, and alignment with contemporary standards in digital health technologies.

From a physical wellbeing perspective, the inclusion of predefined fitness goals during registration, such as weight loss, muscle gain, and flexibility, supports behaviour change by encouraging intentionality and self-regulation. Research shows that digital platforms which incorporate goal-setting mechanisms lead to higher participation and consistency in users' health routines [1]. Although the current prototype does not yet provide personalised plans, its structure prepares the ground for tailored recommendations and AI-driven support in later development phases. This reflects a proactive and professionally sound design strategy that anticipates user needs and aligns with behaviour change models.

Social contribution is also evident in the clear preparation for interactive features, as shown by the navigation placeholders for community elements such as chat, profile pages, and feeds. Online fitness communities are proven to increase motivation, reduce dropout rates, and foster social accountability [2]. By embedding these features into the core navigation at the prototype stage, the project demonstrates foresight and alignment with industry expectations for socially interactive platforms. This approach reflects professionalism by prioritising long-term scalability and user engagement rather than building isolated or non-collaborative features.

Psychological wellbeing is another dimension addressed implicitly in the prototype. Studies indicate that fitness apps which incorporate social connection and supportive environments can help users reduce anxiety and improve self-esteem [3]. The purpose of this study was to examine the influence of the UTAUT2 determinants, as well as the moderating effects of different smartphone fitness app features (ie, education, motivation, and gamification related) and individual differences (ie, age, gender, and experience) on the app usage intentions of individuals and their behavioral intentions of being physically active. The results showed that habit and performance expectancy were the two strongest predictors of intentions of individuals to use fitness apps. The effects of performance expectancy were greater when motivation-related features were rated as important and when education-related features were rated as less important. Moreover, the effects of performance expectancy were greater for males. The effects of habit were greater when education-related features were rated as important and when motivation-related features were rated as less important. Furthermore, the effects of habit were greater for females. Age moderated the relationship between effort expectancy and app usage intention. The intentions of individuals to use fitness apps predicted their intentions of being physically active, using two different means of measuring future physical activity. While the current version does not yet enable interaction, the presence of clear user pathways and motivation-oriented design elements reflects a

professional understanding of user psychology. The neutral colour schemes, uncluttered layouts, and accessible form fields support user confidence and reduce cognitive load.

Inclusivity and accessibility are central to the professionalism demonstrated. The prototype avoids exclusionary design choices by not assuming age, gender, or body type. The language is neutral, input fields are clearly labelled, and navigation is intuitive. This aligns with ethical standards for digital health products, which encourage inclusive design and avoidance of user profiling without justification [4]. Even at the prototype level, avoiding intrusive data collection and using placeholder credentials ensures that development remains GDPR-conscious and ethically consistent.

Professional development practices are also evident in the technical foundations of the prototype. The use of React with modular components ensures maintainability and scalability, while tools like Tailwind or Bootstrap support consistency in UI development. A professional development stack reduces technical debt and enables easier integration of AI features, chat systems, and backend services at later stages [5].

The stringent inclusion criteria are a key strength of this study, positioning it as the first meta-analysis examining the effectiveness of smartphone app–based interventions on objectively measured physical activity. Maturation of this fast-evolving field allowed us to limit the review to randomized controlled trial methodologies and objective outcomes, heightening the trustworthiness of findings and reducing the likelihood of results being influenced by recall or response bias. In addition, an extensive search of 7 databases was undertaken to reduce the risk of publication bias. The navigation bar and structured layout show early planning for integration with authentication systems and real-time communication tools.

In terms of real-world application, STRIVENET positions itself as more than a static app. It is being designed to function as a supportive ecosystem where users can gain physical motivation, psychological reinforcement, and social accountability. Research suggests that platforms which combine fitness guidance with community engagement can significantly improve user outcomes compared to standalone tools . By establishing the visual and structural framework for these outcomes in the prototype, the project shows intentionality and professionalism at the planning stage.

Furthermore, the prototype is suitable for stakeholder demonstration, iteration, and testing. It is visually coherent, functionally representative, and aligns with

the expectations of academic supervisors and industry developers. The absence of real user data, the focus on ethical representation, and the alignment with digital wellbeing trends demonstrate that professionalism is not limited to technical competence but extends to social responsibility.

Overall, the early prototype integrates professionalism by combining ethical considerations, user-centred planning, physical and psychological impact awareness, inclusive design, and scalable architecture. These principles ensure that STRIVENET is not just functional in concept but grounded in real-world value.

2. Social Impact

2.1 Introduction

The social impact of a software artefact refers to how it affects individuals, communities, and society. SOULSTRIVENET is designed to promote **positive behavioural change**, particularly in fitness motivation, consistency, and community engagement.

2.2 Positive Social Impact

SOULSTRIVENET provides several socially beneficial outcomes:

- **Encouraging healthy lifestyles** through AI-driven motivation and personalized fitness guidance.
- **Reducing intimidation barriers** often associated with gyms by offering private AI support.
- **Building community** via a social feed that encourages encouragement and shared progress.

Research shows that AI-supported fitness applications can significantly improve adherence to exercise routines when personalization is present (Patel & Lee, 2022).

2.3 Potential Negative Social Impact

Despite its benefits, potential risks include:

- **Over-reliance on AI advice**, where users may treat FitBot as a replacement for professional trainers.
- **Social comparison pressure** from public posts, which could negatively affect self-esteem.

These risks were mitigated by:

- Avoiding medical claims in AI responses.
- Framing FitBot as a *supportive assistant*, not a certified professional.

2.4 Summary

Overall, SOULSTRIVENET's social impact is predominantly positive, provided users understand the **limitations of AI-generated guidance**.

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3. Ethical Considerations

3.1 Introduction

Ethics in computing concerns fairness, transparency, accountability, and harm prevention. This is especially important for AI-driven systems using **prompt engineering and personalization**.

3.2 AI Ethics and Prompting

FitBot uses **context-enriched and role-based prompting**, which introduces ethical concerns such as:

- Bias amplification
- Over-personalization
- User manipulation

To address this:

- Prompts were designed to **avoid judgmental language**.
- AI responses were constrained to motivational and informational advice.
- Chain-of-thought prompting was used internally but not exposed to users, reducing confusion or false authority (Wei et al., 2022).

3.3 User Wellbeing

Fitness advice affects both **physical and mental wellbeing**. Ethical safeguards included:

- Avoidance of extreme workout recommendations
- Encouraging rest and balance

3.4 Transparency

Users are informed that:

- FitBot is AI-powered
- Responses are generated, not human-authored

This aligns with ethical transparency standards (Floridi et al., 2018).

3.5 Summary

Ethical risks were mitigated through **careful prompt design, transparency, and conservative AI behaviour**, making the system ethically responsible for its scope.

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4. Legal Implications

4.1 Introduction

SOULSTRIVENET operates within the scope of UK and EU digital law, particularly due to its handling of **personal and fitness-related data**.

4.2 GDPR (UK GDPR & EU GDPR)

The project complies with GDPR principles:

- **Lawfulness & Consent:** Users voluntarily provide profile and fitness data.
- **Data Minimisation:** Only necessary fields (e.g., goals, training time) are collected.
- **Purpose Limitation:** Data is used solely for personalization and engagement.
- **Right to Erasure:** Users can log out and request account deletion via Supabase.

Supabase provides GDPR-compliant infrastructure, including data residency options and access controls (Supabase, 2025).

4.3 Health Data Considerations

Although fitness goals are collected, the system avoids medical data, reducing classification as “special category data” under GDPR Article 9.

4.4 Equality Act 2010

The system avoids discriminatory design by:

- Supporting all fitness levels
- Avoiding gendered or ableist assumptions in AI prompts

4.5 Copyright and AI Content

AI-generated content is dynamically produced and not stored permanently, reducing copyright concerns. Users retain ownership of their posts.

4.6 Summary

SOULSTRIVENET demonstrates compliance with relevant UK and EU laws, particularly GDPR and accessibility legislation.

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5. Security Considerations

5.1 Introduction

Security is critical when handling user accounts, personal data, and AI API interactions.

5.2 Authentication and Access Control

Security measures include:

- Supabase-managed authentication
- Token-based session handling
- Protected routes in the frontend

5.3 Data Protection

- All data is transmitted over HTTPS
- Passwords are never stored client-side
- User metadata is sandboxed per account

5.4 AI API Security

- API keys stored in environment variables
- No API secrets exposed in frontend code
- Prompt data sanitized before submission

5.5 User Wellbeing and Safety

To protect users:

- No medical diagnoses provided
- AI avoids extreme or harmful advice
- Error handling prevents misleading outputs

5.6 Summary

The system follows industry-standard security practices appropriate for a university-level AI-driven application.

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6. Conclusion

This companion report demonstrates that SOULSTRIVENET was developed with a strong awareness of **professional responsibility**. Social benefits were prioritised, ethical risks were mitigated through careful AI prompting, legal compliance was respected, and security best practices were implemented.

As the system evolves, future work could include:

- Ethical audits of AI outputs
- Expanded accessibility features
- Enhanced user consent controls

Overall, SOULSTRIVENET reflects a **professionally responsible approach to AI-enhanced software development**, aligned with modern Computer Science standards.

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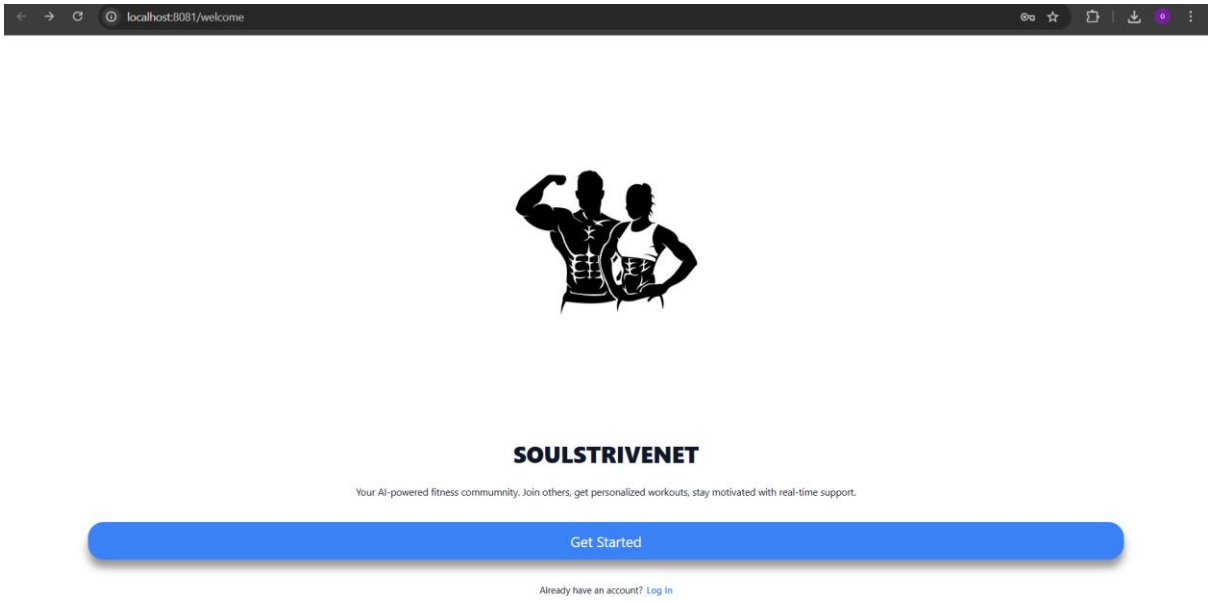
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